

SNAPDRAGON® DOCUMENT ASSISTANT

Offline On-Device Document Intelligence, Plain-Language Simplification & Speech Synthesis for Indian Languages

Target Platform: Snapdragon® X
(Windows 11 ARM64 / HP
OmniBook X)

Architecture: 100% Offline /
Hexagon NPU Offload

Repository:
[https://github.com/openslickofficial/
OpenDoc](https://github.com/openslickofficial/OpenDoc)

1. Executive Summary & Problem Statement

Millions of citizens with limited literacy or second-language English proficiency struggle to understand intimidating official documents—including medical invoices, hospital discharge summaries, utility bills, and municipal legal summonses. Misinterpreting a single payment deadline, claim status, or court date can lead to service disconnection, financial penalties, or loss of critical healthcare benefits. Cloud-based translation services fail in this context: they expose highly private medical and legal records to remote servers, require constant internet connectivity, and routinely hallucinate or alter critical numbers and negation keywords.

Snapdragon Document Assistant (OpenDoc) solves this by providing a completely offline, accessible desktop solution that converts complex paper scans into plain-language summaries and reads them aloud in Indian languages (Hindi, Tamil, Bengali) entirely on the user's local PC, accelerated by the Qualcomm Hexagon NPU.

2. Core 4-Stage On-Device Architecture

- **Stage 1 (Vision OCR):** Ingests document scans, corrects rotational skew via contour bounding, and applies CLAHE contrast normalization. Blank scans short-circuit in under 12 ms.
- **Stage 1 (OCR Extraction):** Runs TrOCR Vision-Encoder-Decoder via ONNX Runtime with Qualcomm QNN Execution Provider, achieving 100% layer placement on the Hexagon Tensor Processor (HTP v73).
- **Stage 2 (Plain-Language Simplification):** Rewrites dense legal/administrative jargon into 6th–8th grade plain English using Qwen3-1.7B via the Qualcomm Genie SDK on NPU.
- **Stage 3 (Indian Language Translation):** Translates simplified content into Indic languages using an Anchor-Preserved Translation Engine with specialized administrative lexicons.
- **Stage 4 (Neural Text-to-Speech):** Synthesizes natural spoken Hindi using on-device Piper TTS ONNX (hi_IN-pratham voice) with zero cloud dependencies.

3. Key Differentiator: Deterministic Fidelity Safeguard

General-purpose generative AI models are notorious for subtle hallucinations: dropping a leading digit (turning \$450.00 into \$50.00), altering dates, or flipping polarity (translating 'Approved' into 'अस्वीकृत' / 'Rejected'). In medical and legal domains, such errors are intolerable.

Safety Decision Engine: Our engine incorporates a strict deterministic cross-language entity verification safeguard. Every financial amount, calendar date, docket ID, and legal status keyword is extracted via exact pattern matching from the source document and verified intact across simplification and Indic translation. If any entity mismatch or contradiction is detected:

- a) Standard Policy (Default):** Prepends an authoritative spoken Hindi warning directly into the synthesized audio: 'चेतावनी: इस दस्तावेज़ में जानकारी की पुष्टि नहीं हो सकी है। कृपया मूल दस्तावेज़ की जाँच करें।' ('Warning: Information in this document could not be fully verified. Please inspect original document.')
- b) Strict Safety Policy:** Completely suppresses audio generation to protect vulnerable users from acting on unverified information.

4. Qualcomm Hexagon NPU Benchmarks & Results

Pipeline Stage	Model / Framework	Hardware Placement	Latency
Document OCR	TrOCR Printed (ONNX)	Hexagon NPU (QNN EP - 100% offload)	1,714 ms
Plain-Language Simplification	Qwen3-1.7B w4a16	Hexagon NPU (Genie SDK / HTP)	1,980 ms
Indic Translation	Anchor-Preserved Lexicon	Local Snapdragon CPU / Genie	4.7 ms
Speech Synthesis	Piper TTS ONNX (hi_IN)	Local CPU / SAPI5 fallback	4,515 ms

5. Security, Privacy & Zero-Network Guarantee

- 100% Offline Runtime: Fully tested and verified with physical network disconnection. Zero cloud APIs, zero telemetry, and zero data leakage.
- Memory-Only Document Processing: Original document scans and extracted PII reside strictly in volatile RAM and are never copied to disk.
- One-Click Privacy Wipe: Integrated '🔪 Clear Session & Cache' button immediately zeroes all volatile memory text buffers and purges all generated speech WAV and JSON artifacts from disk.
- Hardened Input Validation: Implements dimension caps (max 8,000x8,000 px / 32 MP) preventing decompression bombs, handles malformed/corrupted files gracefully in <12 ms, and sanitizes all subprocess calls.